MAHATMA GANDHI UNIVERSITY

III SEMESTER- FYUGP THEORY EXAMINATION 2025

B.Sc BIOLOGICAL TECHNIQUES AND SPECIMEN PREPARATION

MG3DSCBTS200- ESSENTIALS IN BIOLOGICAL TECHNIQUES- DSC A

Time: 1.5hrs **Maximum marks: 50**

[Learning domain][CO NO]

Part A

Fill in the blanks

Answer all questions

Each question carries 1 marks

1) DNA fragments in an agarose gel electrophoresis migrates towards ____**[A] [CO 2,3]**
2) A bioanalytical techniques which is used to separate particles from a solution based on their sedimentation velocity is _____ **[U][CO 2,3]**
3) _____ is an intercalating dye for identifying and visualizing nucleic acid bands in electrophoresis **[K][CO 2,3]**
4) The type of chromatographic technique which exploits the molecular sieve properties of porous material of packing matrix for separating molecules is ____ **[A] [CO 2,3]**
5) _____ is an electric device used to measure hydrogen ion activity in a solution. **[U] [CO2]**
6) A commonly used gel electrophoretic system for separating proteins according to their size is _____ **[U] [CO 2,3]**
7) In agarose gel electrophoresis, smaller the size of DNA fragment their migration will be ___ **[U] [CO 2,3]**
8) _____ is used as a primary fluor in liquid scintillation counting **[U] [CO 6]**
9) _____ tracer is used for localization of calcium in bones **[U] [CO 6]**
10) _____ is an example for soft beta emitter **[k] [CO 6]**

(1x10 =10 marks)

Part B

Short Answer type questions

Answer any 10 out of 12 questions

Each question carries 2 marks

1) State Beer-Lambert's law . Explain its significance in spectrophotometery? **[U][CO 2]**

2) Colorimetry is widely being used for assessing the concentration of target analyte in a solution but they are also known to have certain limitations .Can you cite any 2 merits and demerits of colorimetry? **[U] [CO 2]**
3) What is the basic working principle of centrifugation? **[U] [CO 2,3]**
4) What is Rf value in TLC? **[U] [CO 2,3]**
5) Measuring pH is crucial to determine the acidity or alkalinity of a solution.Write any 3 factors affecting pH determination? **[A] [CO 2]**
6) What are the different types of electrodes used in pH meter ?**[U] [CO 2]**
7) What are the different factors influencing migration rate of DNA molecules in a gel electrophoresis? **[An] [CO 2,3]**
8) AGE is a widely used technique to separate nucleic acids based on size or charge .Why do you think is the reason behind DNA migration through the porous gel of AGE ?**[A] [CO 2,3]**
9) UV –Visible spectroscopy is widely being used in the field of life sciences . Can you cite any 4 applications of UV-Vis spectroscopy? **[A] [CO 2]**
10) What is the application of radiocarbon dating **[U] [CO 6]**
11) Briefly explain bone imaging **[U] [CO 6]**
12) Give four examples for tracers used in biology. **[K] [CO 6]**

(10 x 2 =20 marks)

Part C

Short Essay questions

Answer any 5 out of 7 questions

Each question carries 4 marks

1) Bioanalytics , a field of analytical chemistry for determining metabolites or markers in biological fluids. Describe the importance and the applications of bioanlytics in the field of life sciences with examples ? **[K] [CO 1]**
2) What is the working principle of colorimeter and describe its instrumentation parts with the help of a diagram? **[U] [CO - 2]**
3) What is a DNA ladder ? Explain the significance of using buffers and loading dyes in gel electrophoresis? **[A] [CO 2,3]**
4) Which chromatographic technique would you suggest for separating ionizable molecules based on their charge properties and why? **[U] [CO 2,3]**
5) Cite any 2 types of centrifuges and their application. **[A] [CO 2,3]**
6) Explain in detail autoradiography with applications. **[U] [CO 6]**
7) Illustrate Geiger-Muller counter.**[K] [CO 6]**

(5 x 4 =20 marks)

****